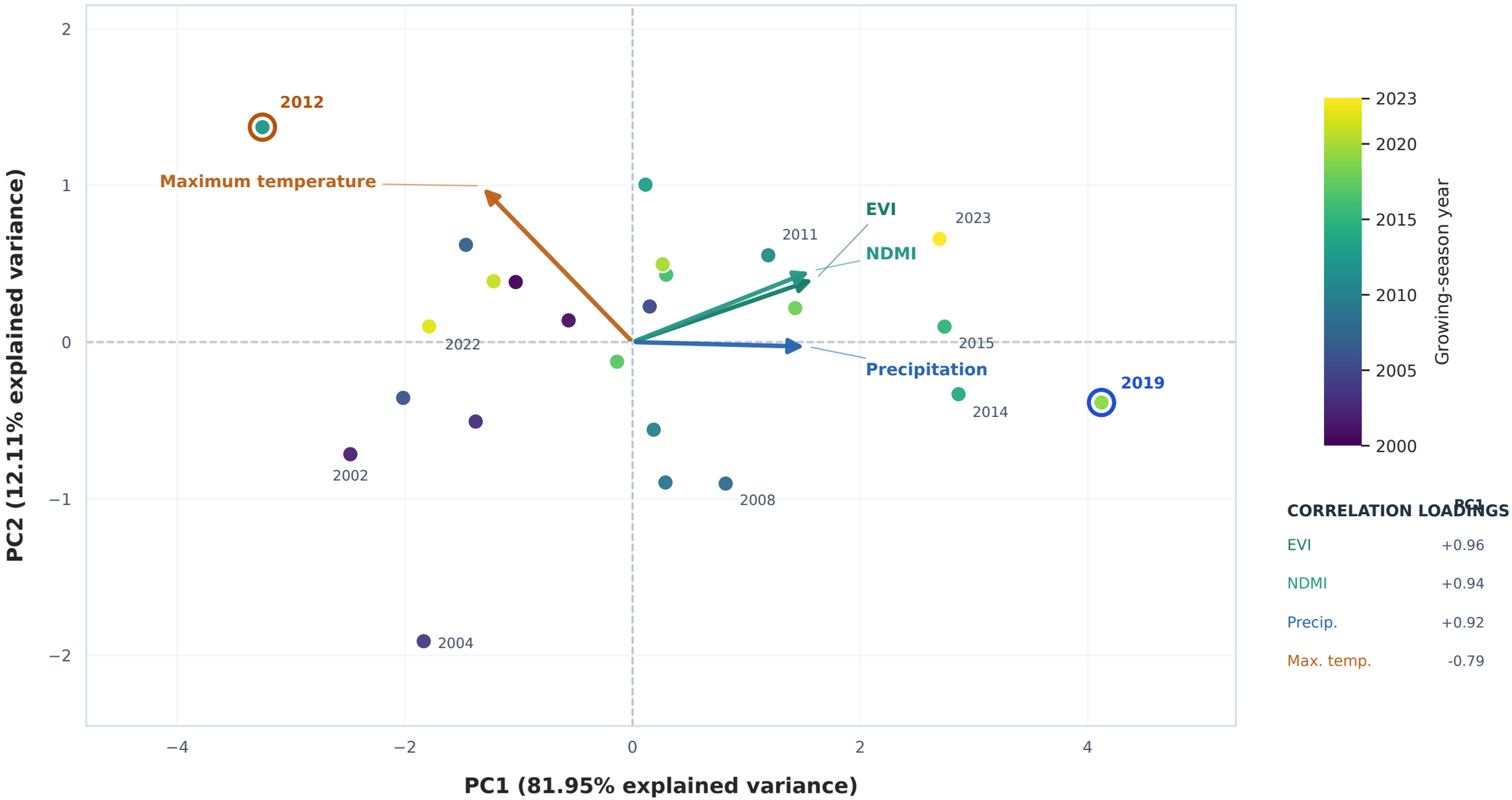
VEGETATION–CLIMATE PRINCIPAL COMPONENT ANALYSIS

Environmental Data Cube for Oglala Lakota

Ate Makoce | May–September 2000–2023 | 24 annual growing seasons

Annual PCA scores and environmental correlation loadings



RESULTS SUMMARY

PC1 explains 81.95% of standardized variance; PC2 explains 12.11% (94.06% combined).

Positive PC1 corresponds to higher vegetation indices and precipitation, and lower maximum temperature.

2012 and 2019 are highlighted as contrasting growing seasons. Only selected years are labeled to improve readability.

Arrows represent correlation loadings rescaled for display. PCA summarizes associations, not causation or grazing impacts.